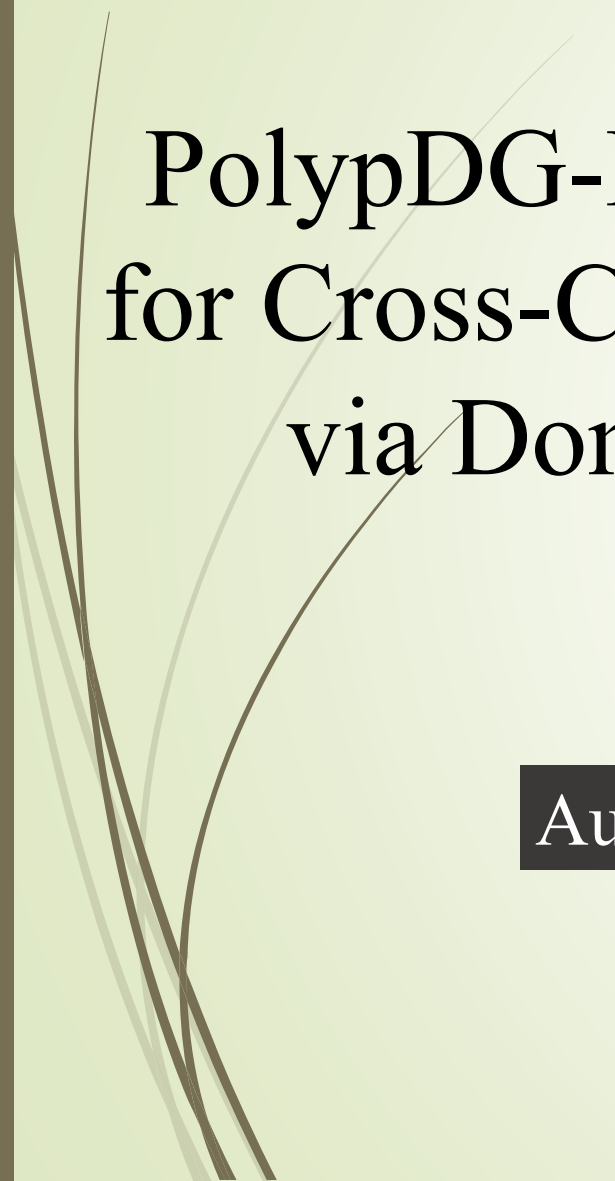




# PolypDG-Lite: A Deployment-Oriented Approach for Cross-Center Colonoscopic Polyp Segmentation via Domain Generalization and Low-Power Knowledge Distillation

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# Outline

**1**

Introduction

**2**

Background

**3**

Preprocessing

**4**

PolypDG-Lite Framework

**5**

Stage 1: Baseline Diagnosis

**6**

Stage 2: Teacher Model Training

**7**

Stage 3: Knowledge Distillation

**8**

Stage 4: Deployment Validation

**9**

Conclusions

# Introduction

## Deep Learning in Colonoscopic Polyp Segmentation



Automated Lesion Localization



Accurate Boundary Delineation



Deployment-Oriented Support

# Background

Random-split evaluation



Overestimated model generalization

Performance-driven models



High development cost



PolypDG-Lite: A deployment-oriented framework for robust cross-center polyp segmentation

# Preprocessing



## Dataset:

PolypGen clean center-labeled subset  
1,537 image–mask pairs from six centers (C1–C6)



## Data Split:

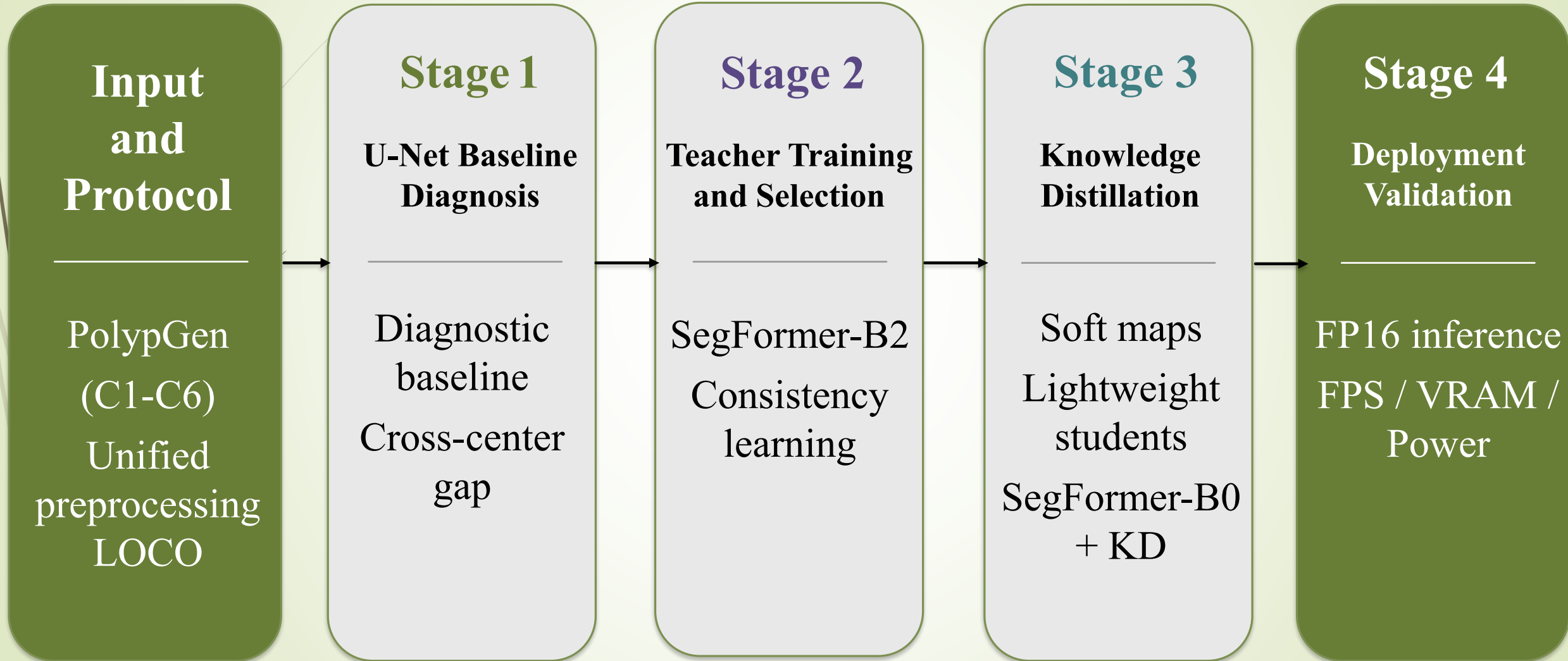
Leave-One-Center-Out (LOCO)  
One center for testing, remaining centers for training and validation



## Preprocessing:

Resize images and masks to  $352 \times 352$   
Normalize image intensity to  $[0, 1]$   
Binarize ground truth masks  
Training: flipping, scale shift, brightness adjustment, contrast jitter

# PolypDG-Lite Framework



# Stage 1: U-Net Baseline Diagnosis

| Center | Dice   | IoU    | HD95  | ASSD  |
|--------|--------|--------|-------|-------|
| C1     | 72.14% | 62.85% | 60.17 | 19.56 |
| C2     | 61.21% | 53.08% | 82.48 | 29.98 |
| C3     | 78.68% | 69.54% | 48.20 | 14.44 |
| C4     | 43.45% | 36.98% | 90.54 | 35.39 |
| C5     | 52.72% | 43.98% | 82.90 | 31.58 |
| C6     | 72.21% | 64.00% | 62.47 | 22.85 |

- Performance varies across centers
- Degraded overlap performance in C4 and C5
- Boundary instability in challenging centers

Baseline performance reveals strong center-dependent variability

# Stage 1: Baseline Generalization Gap

| Fold | Val Dice | $\Delta$ Dice | Val IoU | $\Delta$ IoU |
|------|----------|---------------|---------|--------------|
| C1   | 69.14%   | -3.00%        | 60.48%  | -2.37%       |
| C2   | 66.14%   | 4.92%         | 57.13%  | 4.05%        |
| C3   | 64.12%   | -14.56%       | 56.09%  | -13.44%      |
| C4   | 77.87%   | 34.42%        | 69.08%  | 32.10%       |
| C5   | 69.05%   | 16.33%        | 61.18%  | 17.60%       |
| C6   | 69.58%   | -2.64%        | 60.91%  | -3.09%       |

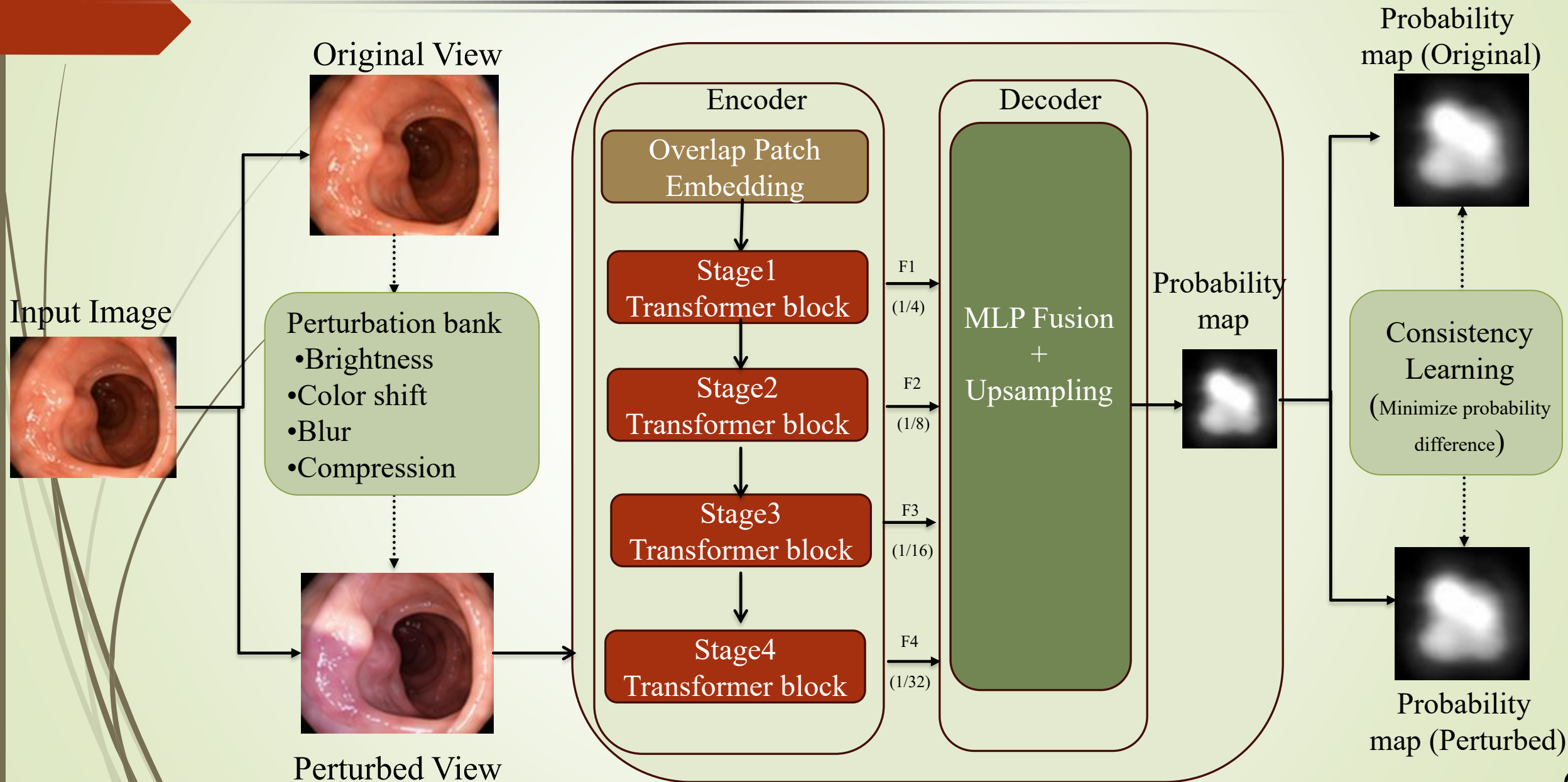
$$\Delta\text{Dice} = \text{Val}_{\text{Dice}} - \text{Test}_{\text{Dice}}$$

$$\Delta\text{IoU} = \text{Val}_{\text{IoU}} - \text{Test}_{\text{IoU}}$$

- Observable generalization gap
- C4 shows the largest degradation
- Smaller absolute  $\Delta\text{Dice}$  and  $\Delta\text{IoU}$  indicate better cross-center stability

Large positive  $\Delta$  values indicate larger performance drop on unseen centers

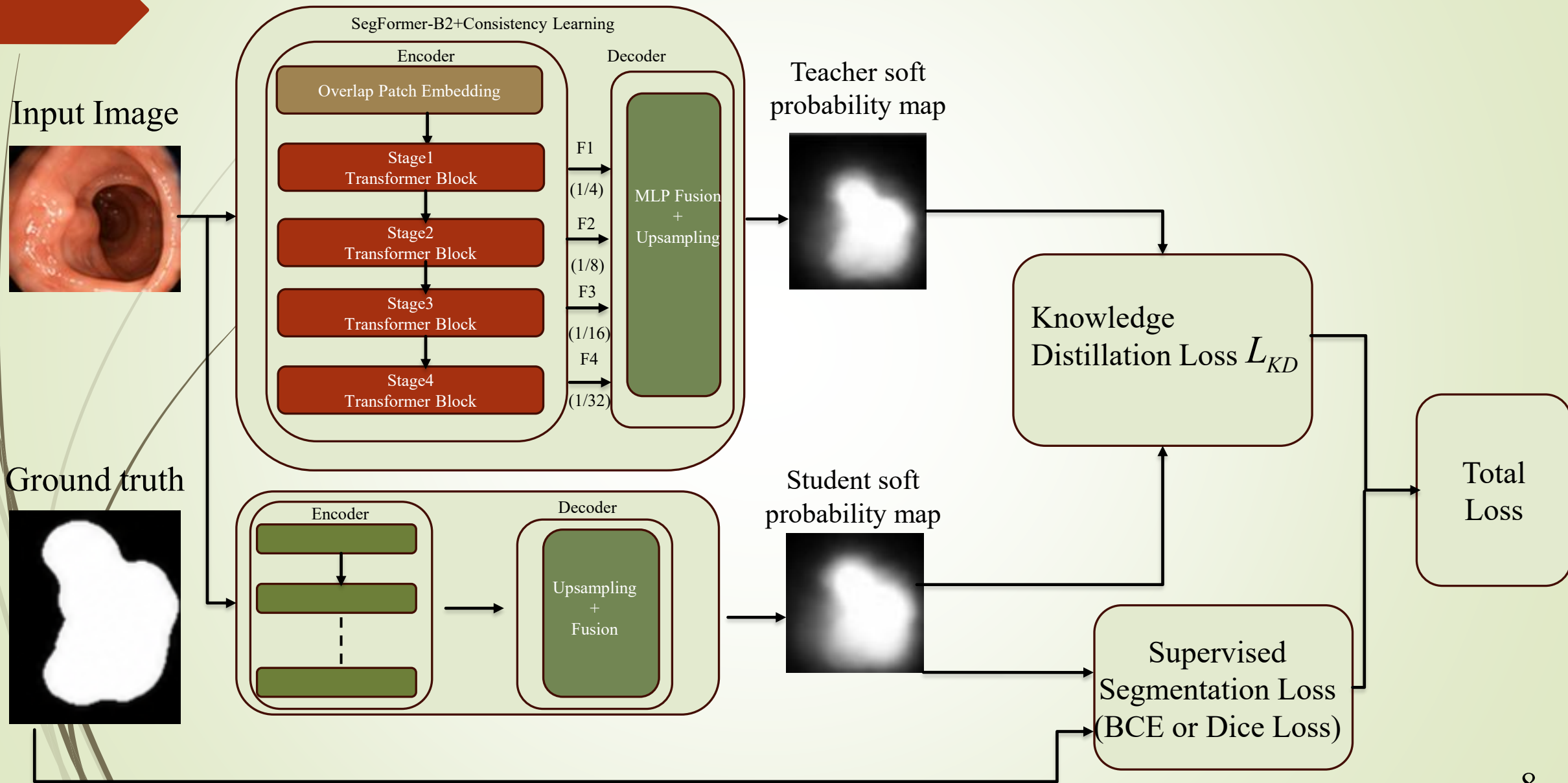
# Stage 2: Teacher Training and Selection



# Teacher Selection Summary

| Method                   | $\Delta Dice$ | $\Delta IoU$ | Worst Fold |
|--------------------------|---------------|--------------|------------|
| SegFormer-B2             | 31.00%        | 29.56%       | C4         |
| +Perturbation Bank       | 32.49%        | 30.56%       | C4         |
| +Consistency<br>Learning | 22.24%        | 20.95%       | C4         |
| +Combined Setting        | 30.56%        | 29.60%       | C4         |

# Stage 3: Knowledge Distillation



# Stage 3 Results

| Student Model     | Dice   | IoU    | Worst Dice | Worst Center | HD95  | ASSD  |
|-------------------|--------|--------|------------|--------------|-------|-------|
| DDRNet-23-slim    | 64.87% | 57.68% | 37.54%     | C4           | 52.89 | 21.22 |
| DDRNet-23-slim+KD | 68.45% | 61.51% | 48.69%     | C4           | 49.47 | 18.64 |
| BiSeNetV2         | 64.06% | 56.88% | 41.94%     | C4           | 52.39 | 20.24 |
| BiSeNetV2+KD      | 65.21% | 58.44% | 41.69%     | C4           | 49.21 | 18.68 |
| SegFormer-B0      | 73.15% | 66.15% | 47.02%     | C4           | 36.43 | 12.42 |
| SegFormer-B0+KD   | 77.05% | 70.38% | 60.44%     | C4           | 32.81 | 10.63 |

| Student Model     | Dice(Val.) | $\Delta$ Dice | IoU(Val.) | $\Delta$ IoU |
|-------------------|------------|---------------|-----------|--------------|
| DDRNet-23-slim+KD | 79.91%     | 31.22%        | 72.65%    | 28.57%       |
| BiSeNetV2+KD      | 79.48%     | 37.79%        | 71.74%    | 35.13%       |
| SegFormer-B0+KD   | 85.24%     | 24.80%        | 78.56%    | 23.06%       |

# Stage 4: Deployment Validation

## Deployment Setting

- Select model: SegFormer-B0+Knowledge Distillation
- Comparison

FP32: standard inference precision

FP16: reduced precision for efficient deployment

| Model           | Precision | FPS   | Latency(ms) | VRAM(MB) | Power(W) | J/frame |
|-----------------|-----------|-------|-------------|----------|----------|---------|
| SegFormer-B2    | FP32      | 31.44 | 32.08       | 422.35   | 62.61    | 1.918   |
| SegFormer-B0+KD | FP32      | 87.05 | 11.57       | 125.64   | 26.49    | 0.546   |
| SegFormer-B0+KD | FP16      | 95.26 | 10.53       | 55.43    | 14.77    | 0.382   |

# Conclusions

## Cross-Center Generalization

LOCO reveals clear cross-center generalization gaps under domain shift.

## Key Insights

The staged teacher–student framework improves robustness and deployment efficiency across key evaluation metrics.

## Best Performing Model

SegFormer-B0+KD with FP16 deployment

Segmentation: Dice: 77.05%, IoU: 70.38%,

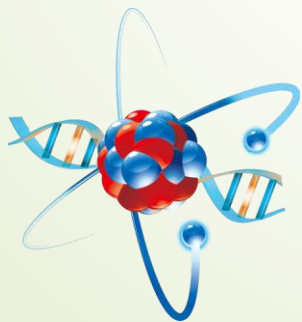
Efficiency: FPS: 95.26, Energy: 0.382J/frame

## Future Work

Broader multi-center validation and real-time clinical deployment.



Thanks for your attention and  
participation



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